UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Particular Integral, Type 3: Polynomial Right-Hand Side

Complete study pack for class 005: the rule $\frac{1}{f(D)}x^m = [f(D)]^{-1}x^m$ (take the lowest-degree term out, expand $[1 \pm \varphi(D)]^{-1}$ by the binomial theorem up to D^m, operate term by term), the four standard expansions, the case where D is a factor of $f(D)$, and the ESA questions that mix Types 1, 2 and 3 (e^{ax}, a^x, $\sin$, $\cos$ and a polynomial) by superposition

Compiled from the PESU Unit 2 class 005 slide deck, its AV-summary transcript (T005), the Type 3 part of the class 006 problem lecture (T006a), the Unit 1 & 2 class-work sheet (problems 15, 16, 21, 22) and eight past ESA papers. Generated 16 October 2026.

· How to use this pack

Topic 1 is the Type 3 rule as the slides state it (Figure F1.1): take the lowest-degree term out, invert, expand, differentiate. Topic 2 is the expansion: the four series of slide 005·5 and where to stop (Figure G2.1). Topic 3 is $f(D)$ with a factor D (added for clarity). Topic 4 is the exam core: polynomials mixed with e^{ax}, a^x, $\sin$, $\cos$, solved by superposition (Figures F4.1–F4.6). CF cases: class 002 pack; Type 1: class 003; Type 2: class 004. They are not re-taught here, but every worked example is complete. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·8	Unit 2 deck 005 (“Particular integral of Type 3”), slide 8 (rendered .pptx)
T005 00:06:48	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 3” (12.6 min), segment start
T006a 00:06:05	Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min), its Type 3 part 00:00:00–00:09:55; the class 006 pack cites the whole file as T006b
CW U2 P15	Unit 2 class-work problem 15
ESA Jul-23 3b	Past paper: UE22MA141A Jul 2023, question 3.b
ESA Jan-24 s2 2d	Past paper: UE23MA141A Jan 2024 set 2, question 2.d

Contents

1. 1 · The Type 3 rule: $X = x^m$
2. 2 · The binomial expansions and where to stop
3. 3 · When D is a factor of $f(D)$
4. 4 · Mixed right-hand sides: Types 1, 2 and 3 together
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 The Type 3 rule: $X = x^m$

When the right-hand side is a polynomial (x^m, or a sum of powers such as $x^2 + 7x + 9$), the PI $\frac{1}{f(D)}x^m$ is found by writing $\frac{1}{f(D)}$ as a series in ascending powers of D. Take the lowest-degree term of $f(D)$ out, so that $f(D) = a_0[1 \pm \varphi(D)]$; move the bracket to the numerator as $[1 \pm \varphi(D)]^{-1}$; expand it by the binomial theorem; and let each power of D differentiate the polynomial. The series can stop at D^m, because every higher derivative of x^m is zero.

Key terms and definitions

Type 3 right-hand side. $X = x^m$, m a positive integer; by superposition, any polynomial in x. S005·3 · S005·4 · S005·10 · T005 00:00:18

Lowest-degree term. The term of $f(D)$ with the smallest power of D: the constant $a_0 = a_0D^0$ when there is one ("minus 1 can be written as minus 1 into D to the power 0"), otherwise D or D^k. S005·4 · T005 00:06:01

$\varphi(D)$. What is left of $f(D)$ after the lowest-degree term is taken out: $f(D) = a_0[1 \pm \varphi(D)]$. It contains only positive powers of D. S005·4 · T005 00:01:08

> *"Type (iii); When $X = x^n$, 'n' being a positive integer. Here PI $= \frac{1}{f(D)}(x^n)$. Take the lowest degree term common from $f(D)$ to get an expression in the form $(1 \pm \varphi(D))$ in the denominator. Take this factor to the numerator to get $(1 \pm \varphi(D))^{-1}$. Expand $(1 \pm \varphi(D))^{-1}$ using Binomial theorem upto nth power as $(n+1)^{\text{th}}$ derivative of x^n is zero. Operate on the numerator term by term by taking $D = \frac{d}{dx}$."* S005·4

> *"If $X = x^m$, then the particular integral is $y_p = \text{P.I} = \frac{1}{f(D)}x^m = [f(D)]^{-1}x^m$. Expand $[f(D)]^{-1}x^m$ in ascending powers of D using binomial theorem and operate on x^m term by term. Since $(m+1)$th term and higher derivatives are zero, it is not necessary to consider terms beyond D^m."* S005·10

Formulas

E1.1 · Type 3 PI ESA ★★★

$$y_p = \text{P.I} = \frac{1}{f(D)}x^m = \big[f(D)\big]^{-1}x^m$$

gives	The PI of a power of x (or a polynomial).
applies	$X = x^m$, m a positive integer.
note	The PI is a polynomial of degree m when $f(0) \neq 0$.

S005·4 · S005·10 · T005 00:00:50 · T006a 00:00:36

E1.2 · Take the lowest-degree term out ESA ★★★

$$\frac{1}{f(D)}x^m = \frac{1}{a_0\,[1 \pm \varphi(D)]}x^m = \frac{1}{a_0}\big[1 \pm \varphi(D)\big]^{-1}x^m$$

GIVES A bracket ready for the binomial series.

APPLIES a_0 = the lowest-degree term (usually the constant).

NOTE Take out the constant with its sign: $D^2 - 1 = -(1 - D^2)$, $D^2 + 5D + 4 = 4\big[1 + \frac{D^2+5D}{4}\big]$.

S005·4 · S005·7 · S005·8 · T005 00:01:08 · T005 00:09:16

E1.3 · Powers of D on x^m ESA ★★★

$$D^k x^m = \frac{m!}{(m-k)!}\,x^{m-k}\ (k \le m), \qquad D^k x^m = 0\ (k > m)$$

GIVES Why the series stops at D^m; $D^m x^m = m!$ is a constant.

APPLIES Operating term by term, $D = \frac{d}{dx}$.

NOTE Example: $D(x^2 + 7x + 9) = 2x + 7$, $D^2(x^2 + 7x + 9) = 2$, $D^3(\dots) = 0$.

S005·4 · S005·10 · T005 00:01:51 · T006a 00:01:29 · T006a 00:02:06

Graphs and diagrams

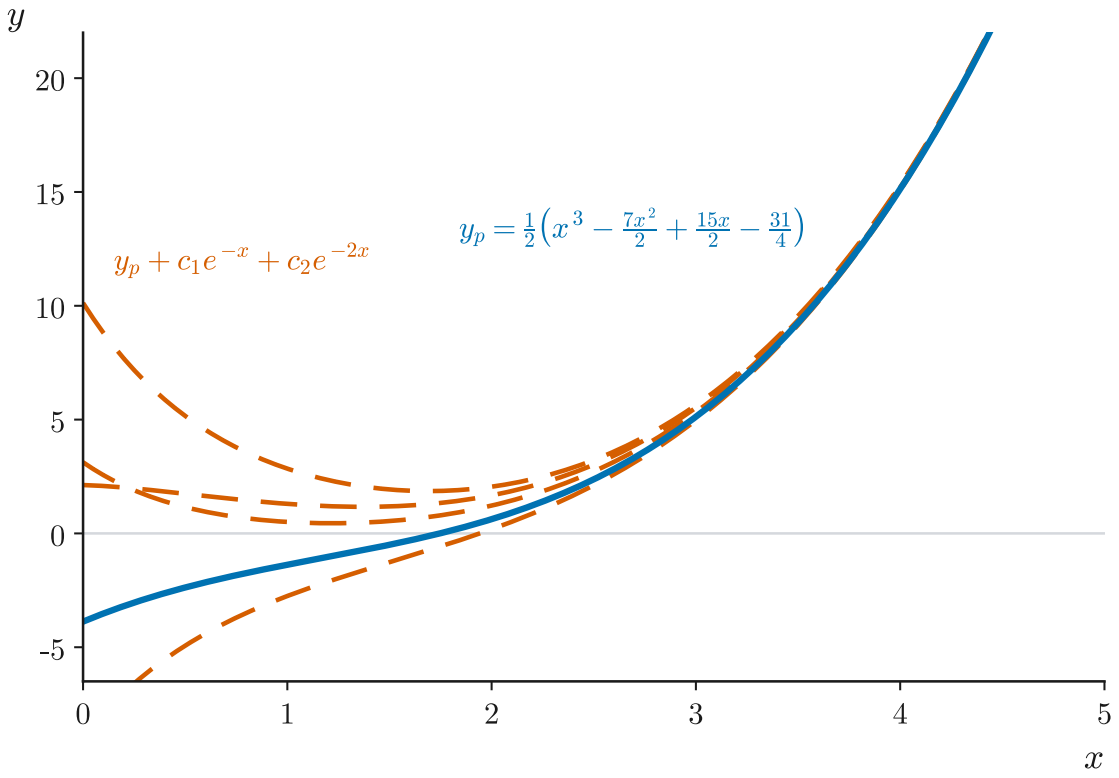
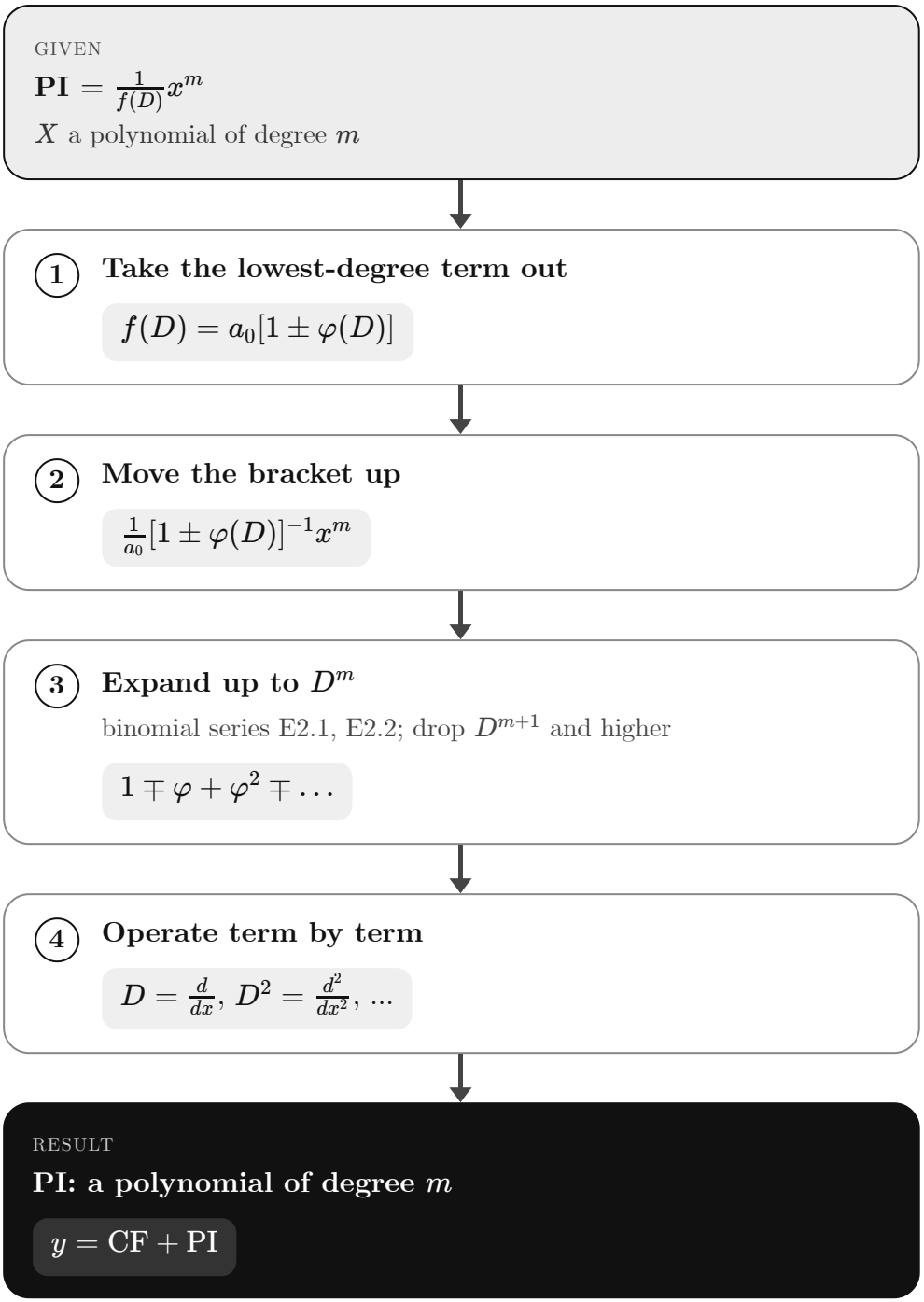


Figure G1.1. Every solution of $(D^2 + 3D + 2)y = x^3 + x^2$ settles onto the PI. ESA ★★

ADDED FOR CLARITY S005·11 · S005·12

How to read it. The PI $y_p = \frac{1}{2}(x^3 - \frac{7x^2}{2} + \frac{15x}{2} - \frac{31}{4})$ (blue) is a cubic, the same degree as X. Four general solutions $y_p + c_1 e^{-x} + c_2 e^{-2x}$ (vermilion dashed) start at different values but the CF terms die out (roots -1, -2), so they all merge with y_p within a few units of x.

Flowcharts



Remark. Collect the operator $(1 + c_1D + \cdots + c_mD^m)$ before differentiating.

Exam trap. Keep the factor $\frac{1}{a_0}$ and its sign ($D^2 - 1 = -(1 - D^2)$).

Figure F1.1. The Type 3 procedure (slide 005·4) ESA ★★★

S005·4 · S005·10 · T005 00:01:08 · T005 00:02:17 · T006a 00:00:55

Types 1, 2 and 3 side by side

	Type 1: e^{ax}	**Type 2:** $\sin ax$, $\cos ax$	**Type 3:** x^m
What to do with D	replace D by a	replace D^2 by $-a^2$	expand $[f(D)]^{-1}$ in powers of D
PI	$\frac{e^{ax}}{f(a)}$	$\frac{\sin ax}{f(-a^2)}$	$\frac{1}{a_0}[1 \pm \varphi(D)]^{-1}x^m$
Special case	$f(a)=0$: $x\frac{e^{ax}}{f'(a)}$	$f(-a^2)=0$: $x\frac{\sin ax}{f'(-a^2)}$	$f(0)=0$: take D^k out, integrate
PI looks like	a multiple of e^{ax}	$A\cos ax + B\sin ax$	a polynomial of degree m (or $m+k$)
Class pack	003	004	005 (this pack)

ADDED FOR CLARITY S005·4 · S005·10

Important points and why

1. Recognise Type 3: the right-hand side is an algebraic function, a polynomial in x ("$5x-2$", "x^2+7x+9"). ESA ★★★ T005 00:04:22 · S005·3
2. Take out the **lowest**-degree term of $f(D)$, with its sign and coefficient: $D^2-1=-(1-D^2)$, $2D^2+2D+3=3\left[1+\frac{2D^2+2D}{3}\right]$. ESA ★★★ S005·4 · S005·7 · S005·13 · T005 00:06:01 · T006a 00:08:22
3. Move $[1 \pm \varphi(D)]$ to the numerator as $[1 \pm \varphi(D)]^{-1}$; the constant $\frac{1}{a_0}$ stays in front. ESA ★★★ S005·4 · T005 00:01:21 · T005 00:09:27
4. Expand only up to D^m, where m is the degree of the polynomial (not the order of the equation): "the $(m+1)$th and higher derivatives are zero". ESA ★★★ S005·4 · S005·10 · T005 00:01:51 · T006a 00:05:30
5. Operate term by term: D differentiates once, D^2 twice, D^3 three times. ESA ★★★ S005·4 · T005 00:02:17 · T006a 00:07:02
6. If the question says "find the PI", the CF is not needed (slide 005·8); "solve" or "general solution" needs $y=y_c+y_p$. ESA ★★ S005·8 · T005 00:08:25 · S005·7
7. When $f(0)\neq 0$ the PI is a polynomial of the same degree as X; check the leading coefficient: it is the leading coefficient of X divided by $f(0)$. ESA ★★ ADDED FOR CLARITY S005·9 · S005·12

Why take out the lowest-degree term, not the highest?
Then $\varphi(D)$ has only positive powers of D, and the series is in ascending powers: each term is a derivative, and all but finitely many vanish on x^m. Taking out the highest power would give a series in $\frac{1}{D}$ (repeated integrals) that never ends.

ADDED FOR CLARITY S005·4 · T005 00:01:08

Why may the infinite series be cut at D^m?
$D^{m+1}x^m = 0$, and so is every higher derivative: "D^2, D^4, D^6 of $5x-2$ is 0". The terms dropped contribute nothing, so the truncated operator gives the exact PI, not an approximation.
S005·4 · T005 00:07:49 · T006a 00:01:29

Why does a power series in D give a valid PI at all?
On polynomials of degree m, $f(D)$ acts like a polynomial in D with $D^{m+1}=0$, so the truncated series is an exact inverse: $f(D)\,[\text{series}]\,x^m = x^m$. Substituting the PI back always returns X (checked with sympy for every example here).
ADDED FOR CLARITY S005·10

Exam trap. Slide 005·13: the roots of $2m^2+2m+3=0$ are $\frac{-1\pm\sqrt{5}i}{2}$, not $\frac{-2\pm\sqrt{5}i}{2}$; slides 005·13–14 also drop c_2 in the CF. ADDED FOR CLARITY S005·13 · S005·14 · T006a 00:07:45

Exam trap. Do not forget the factor $\frac{1}{a_0}$ and its sign: taking -1 out of D^2-1 puts a minus in front of the whole PI. S005·7 · T005 00:06:17

Exam trap. Stop at D^m where m is the degree of X, not the order of $f(D)$: for x^5 and D^3-1 you need $1+D^3$ (up to D^5), not just 1. ADDED FOR CLARITY CW U2 P16 · S005·10

Exam trap. T006a 00:02:23 says the nth derivative of x^m is zero "when n is m"; it is zero for $n>m$ ($D^m x^m = m!$). ADDED FOR CLARITY T006a 00:02:23 · T006a 00:01:29

Worked examples

W1.1 · $\frac{d^2y}{dx^2} - y = 5x-2$ (slides 005·6–7) ESA ★★★

GIVEN Find the general solution of $\frac{d^2y}{dx^2} - y = 5x-2$, i.e. $(D^2-1)y = 5x-2$.

STEP 1 AE $m^2-1=0$: $m=\pm1$. CF $= c_1e^x + c_2e^{-x}$.

STEP 2 PI $= \frac{5x-2}{D^2-1} = -\frac{5x-2}{1-D^2} = -(1-D^2)^{-1}(5x-2)$ (lowest-degree term -1 taken out).

STEP 3 $(1-z)^{-1} = 1+z+\ldots$: $-(1+D^2+D^4+\ldots)(5x-2)$; $m=1$, and $D^2(5x-2)=0$.

STEP 4 PI $= -(5x-2) = 2-5x$.

ANSWER $\boxed{y = c_1e^x + c_2e^{-x} + 2 - 5x}$

CHECK sympy dsolve and checkodesol agree. The slide expands with the wrong signs $(1-D^2+D^4)$; the answer is unaffected.

S005·6 · S005·7 · T005 00:04:22 · T005 00:08:07

W1.2 · PI of $(D^2+5D+4)y = x^2+7x+9$ (slides 005·8–9) ESA ★★★

GIVEN Find the particular integral of $(D^2+5D+4)y = x^2+7x+9$.

STEP 1 $\text{PI} = \frac{x^2+7x+9}{4\left[1+\frac{1}{4}(D^2+5D)\right]} = \frac{1}{4}\left[1+\frac{1}{4}(D^2+5D)\right]^{-1}(x^2+7x+9)$.

STEP 2 Up to D^2: $1-\frac{1}{4}(D^2+5D)+\frac{1}{16}(D^2+5D)^2 \to 1-\frac{D^2}{4}-\frac{5D}{4}+\frac{25D^2}{16}$ (drop D^4 and $10D^3$).

STEP 3 $D(x^2+7x+9) = 2x+7$, $D^2(\dots) = 2$: $\frac{1}{4}\left[x^2+7x+9-\frac{2}{4}-\frac{5(2x+7)}{4}+\frac{25\cdot 2}{16}\right]$.

ANSWER $$\boxed{\text{PI} = \frac{1}{4}\left(x^2+\frac{9}{2}x+\frac{23}{8}\right)}$$

CHECK Slide answer agrees; substituting back gives x^2+7x+9 (sympy). With CF $c_1e^{-x}+c_2e^{-4x}$ this is the general solution.

S005·8 · S005·9 · T005 00:08:25 · T005 00:12:07

W1.3 · $(D^2-2D-3)y = 2x^2+6x$ (CW P15) ESA ★★

GIVEN Solve $(D^2-2D-3)y = 2x^2+6x$.

STEP 1 AE $(m-3)(m+1)=0$: CF $= c_1e^{-x}+c_2e^{3x}$.

STEP 2 $D^2-2D-3 = -3\left[1+\frac{2D-D^2}{3}\right]$, so $\text{PI} = -\frac{1}{3}\left[1+\frac{2D-D^2}{3}\right]^{-1}(2x^2+6x)$.

STEP 3 Up to D^2: $1-\frac{2D-D^2}{3}+\frac{(2D-D^2)^2}{9} \to 1-\frac{2D}{3}+\frac{D^2}{3}+\frac{4D^2}{9} = 1-\frac{2D}{3}+\frac{7D^2}{9}$.

STEP 4 $D(2x^2+6x) = 4x+6$, $D^2 = 4$: $-\frac{1}{3}\left[2x^2+6x-\frac{8x+12}{3}+\frac{28}{9}\right] = -\frac{1}{3}\left[2x^2+\frac{10x}{3}-\frac{8}{9}\right]$.

ANSWER $$\boxed{y = c_1e^{-x}+c_2e^{3x}-\frac{1}{27}(18x^2+30x-8)}$$

CHECK CW answer agrees; sympy dsolve and checkodesol agree.

CW U2 P15

SHORT-NOTE RECAP

Type 3: $X = x^m$. $\text{PI} = \frac{1}{f(D)}x^m = [f(D)]^{-1}x^m$. Take the lowest-degree term out, $f(D) = a_0[1\pm\varphi(D)]$; write $\frac{1}{a_0}[1\pm\varphi(D)]^{-1}$; expand by the binomial theorem up to D^m (since $D^{m+1}x^m = 0$); operate term by term with $D = \frac{d}{dx}$. Examples: $\frac{5x-2}{D^2-1} = 2-5x$; $\frac{x^2+7x+9}{D^2+5D+4} = \frac{1}{4}(x^2+\frac{9}{2}x+\frac{23}{8})$; CW P15: $\frac{2x^2+6x}{D^2-2D-3} = -\frac{18x^2+30x-8}{27}$.

2 The binomial expansions and where to stop

Four series do all the work: $(1+z)^{-1}$, $(1-z)^{-1}$, $(1+z)^{-2}$ and $(1-z)^{-2}$, with $z = \varphi(D)$. Only the first and third need to be remembered; the other two follow by replacing z with $-z$. When $\varphi(D)$ has two terms, its square and cube contain many products, but only those with a power of D at most m survive. Collect the operator first, then apply it to the polynomial.

Key terms and definitions

Ascending powers of D. A series $1 + c_1D + c_2D^2 + \ldots$; applied to x^m it ends after D^m. S005·5 · S005·10 · T005 00:01:39

Truncation. Dropping every term of the expansion whose power of D exceeds m, since $D^{m+1}x^m = 0$. S005·4 · S005·10 · T006a 00:01:15

> *"The following expressions will be useful to expand $(1 \pm \varphi(D))^{-1}$ in ascending powers of D."* S005·5

Formulas

E2.1 · First-power series ESA ★★★

$$(1+z)^{-1} = 1 - z + z^2 - z^3 + \ldots$$
$$(1-z)^{-1} = 1 + z + z^2 + z^3 + \ldots$$

gives	$[1 \pm \varphi(D)]^{-1}$ with $z = \varphi(D)$.
applies	After E1.2.
note	Signs alternate for $(1+z)^{-1}$; all plus for $(1-z)^{-1}$ (replace z by $-z$).

S005·5 · T005 00:02:45 · T005 00:03:12 · T006a 00:04:36

E2.2 · Second-power series ESA ★★

$$(1+z)^{-2} = 1 - 2z + 3z^2 - 4z^3 + \ldots$$
$$(1-z)^{-2} = 1 + 2z + 3z^2 + 4z^3 + \ldots$$

gives	$[1 \pm \varphi(D)]^{-2}$ when $f(D)$ is a square, e.g. $(D-2)^2$, $(D^2-1)^2$.
applies	$f(D) = a_0^2[1 \pm \varphi(D)]^2$.
note	"If we remember the first and the third, it is sufficient."

S005·5 · T005 00:03:46 · T005 00:04:04 · CW U2 P21 · CW U2 P22

E2.3 · Keep only powers up to D^m ESA ★★★

$$(D^2+3D)^2 = D^4+6D^3+9D^2 \;\to\; 6D^3+9D^2$$
$$(D^2+3D)^3 = D^6+\cdots+27D^3 \;\to\; 27D^3 \qquad (m=3)$$

GIVES The terms of the square and cube that survive.

APPLIES $\varphi(D)$ has two or more terms.

NOTE Lecture: "we don't want a^2, we want b^2, ... $2ab$ that is $6D^3$".

S005·12 · T006a 00:06:05 · T006a 00:06:22 · T006a 00:06:43 · T005 00:11:24

Graphs and diagrams

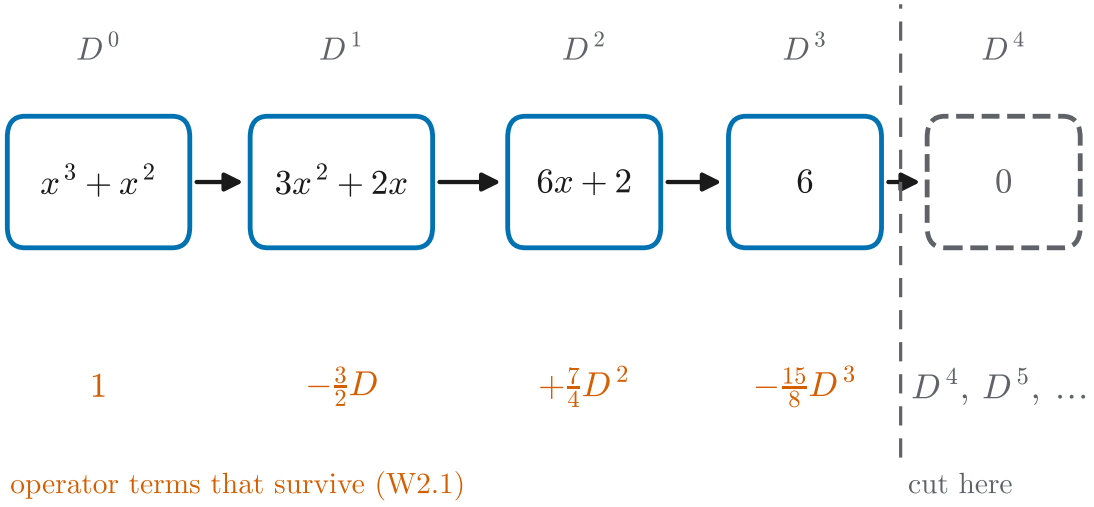
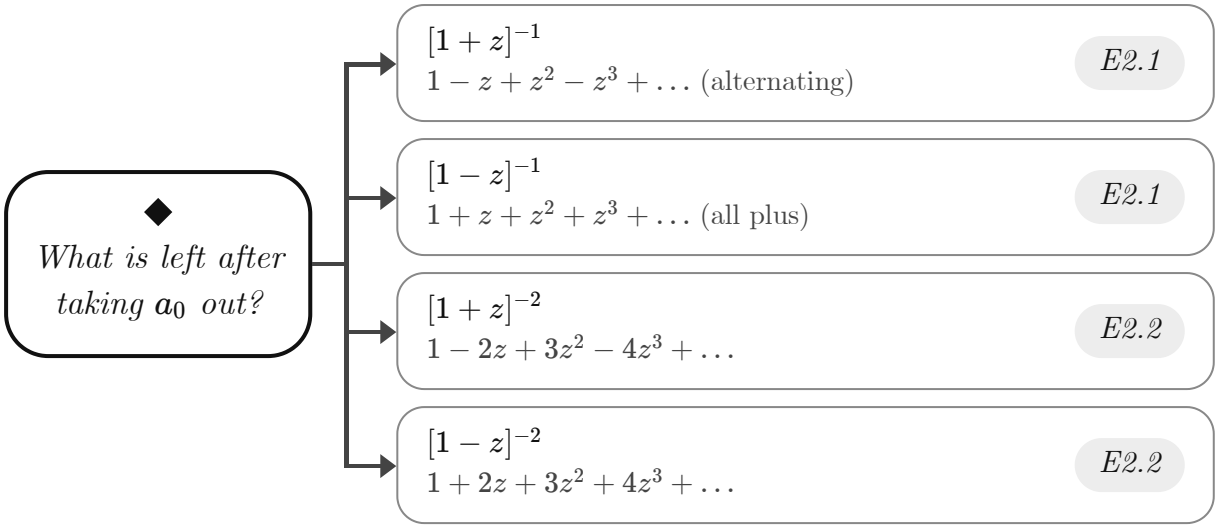


Figure G2.1. Why the series stops at D^m: the derivative staircase of x^3+x^2. ESA ★★★

ADDED FOR CLARITY S005·4 · S005·10 · S005·12 · T006a 00:05:45

How to read it. Each step to the right is one more D. After three steps the polynomial is the constant 6, and D^4 gives 0 (grey). So in $[1+\frac{D^2+3D}{2}]^{-1}$ only the operator terms $1, D, D^2, D^3$ (shown under the boxes with their coefficients from W2.1) can contribute; D^4, D^5, D^6 are dropped (schematic).

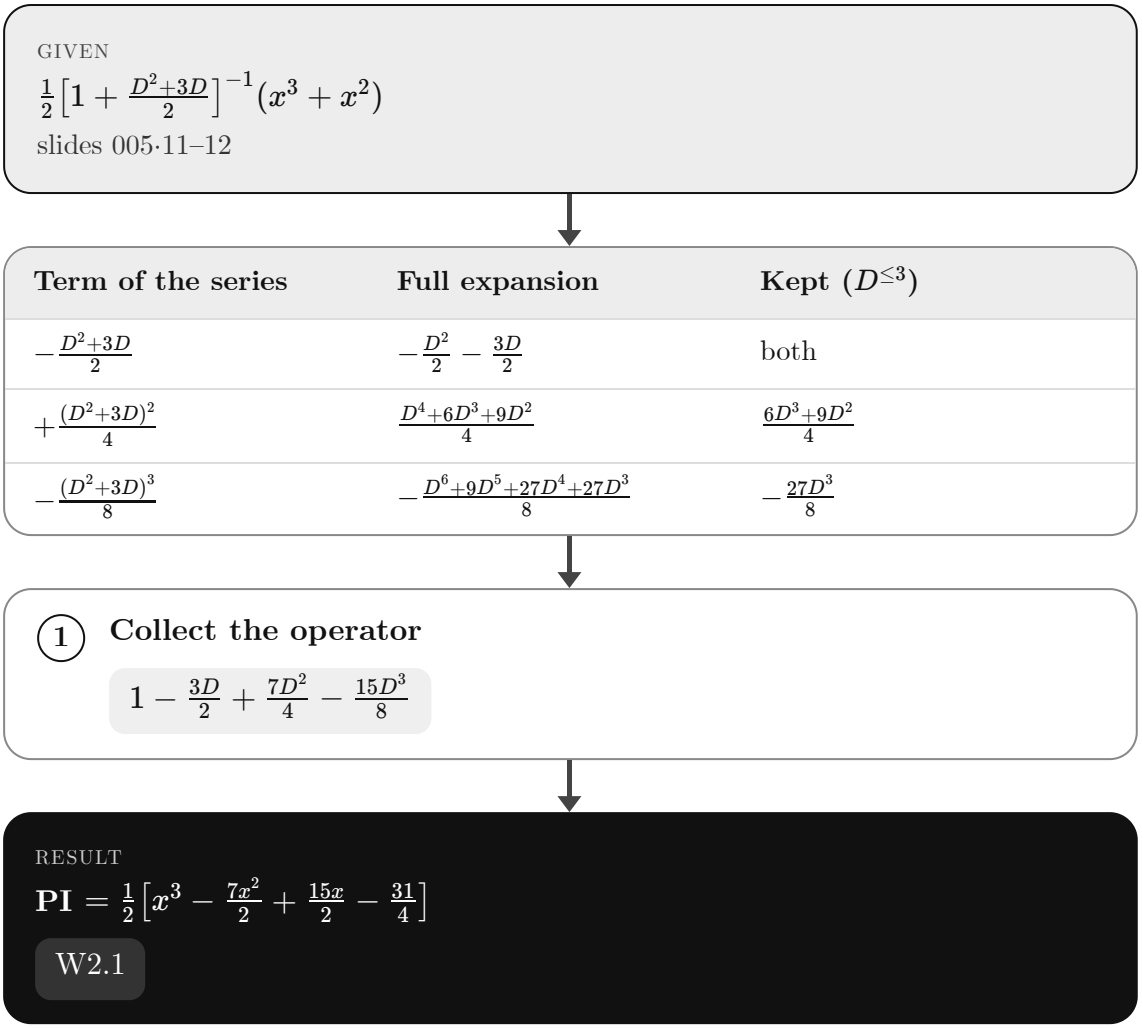
Flowcharts



Remark. $z = \varphi(D)$. Remember the first and third rows; the others follow by $z \to -z$.

Figure F2.1. Which series? ESA ★★★

S005·5 · T005 00:02:45 · T005 00:04:04



Remark. The fourth power and beyond are not written at all: every product in them has D^4 or more.

Figure F2.2. Which products survive? (x^3+x^2, so $m=3$) ESA ★★★

S005·12 · T006a 00:06:05 · T006a 00:06:43

The four expansions of slide 005·5

Expression	Series	Signs	Remember?
$(1+z)^{-1}$	$1-z+z^2-z^3+\dots$	alternate	yes
$(1-z)^{-1}$	$1+z+z^2+z^3+\dots$	all plus	from row 1, $z\to -z$
$(1+z)^{-2}$	$1-2z+3z^2-4z^3+\dots$	alternate	yes
$(1-z)^{-2}$	$1+2z+3z^2+4z^3+\dots$	all plus	from row 3, $z\to -z$

S005·5 · T005 00:03:03 · T005 00:04:04

Important points and why

1. Remember two series: $(1+z)^{-1} = 1 - z + z^2 - \dots$ and $(1+z)^{-2} = 1 - 2z + 3z^2 - \dots$; get the other two by $z \to -z$ (all signs plus). ESA ★★★ S005·5 · T005 00:03:12 · T005 00:04:04
2. Expand the square or cube of a two-term $\varphi(D)$ by $(a+b)^2$, $(a+b)^3$, but write down only the products with D-power $\le m$. ESA ★★★ S005·9 · S005·12 · T005 00:10:49 · T006a 00:06:05
3. Collect the operator into $1 + c_1 D + c_2 D^2 + \dots + c_m D^m$ before applying it: fewer derivatives, fewer slips. ESA ★★ (added for clarity) S005·9 · T006a 00:06:43
4. For $f(D) = (D-a)^2$ or $(D^2-1)^2$ use the $(1 \pm z)^{-2}$ series directly, instead of squaring the first-power series. ESA ★★ S005·5 · CW U2 P21 · CW U2 P22

Exam trap. Slide 005·7 writes $-(1-D^2)^{-1} = -(1 - D^2 + D^4 - \dots)$; it is $-(1 + D^2 + D^4 + \dots)$ (the lecturer corrects it, T005 00:07:07). The answer $2 - 5x$ survives only because $D^2(5x - 2) = 0$. (added for clarity) S005·7 · T005 00:07:07

Exam trap. In $(D^2 + 3D)^2$ the term $6D^3$ is kept when $m = 3$ but dropped when $m = 2$; decide product by product. T006a 00:06:22 · T005 00:11:24

Worked examples

W2.1 · $(D^2 + 3D + 2)y = x^3 + x^2$ (slides 005·11–12) ESA ★★★

given Solve $(D^2 + 3D + 2)y = x^3 + x^2$.

step 1 AE $m^2 + 3m + 2 = 0$: $m = -1, -2$. CF $= c_1 e^{-x} + c_2 e^{-2x}$.

step 2 PI $= \frac{1}{2}\left[1 + \frac{D^2+3D}{2}\right]^{-1}(x^3 + x^2)$; expand up to D^3 ($m = 3$).

step 3 $1 - \frac{D^2+3D}{2} + \frac{(D^2+3D)^2}{4} - \frac{(D^2+3D)^3}{8}$; keep $9D^2 + 6D^3$ of the square and $27D^3$ of the cube: $1 - \frac{3D}{2} + \frac{7D^2}{4} - \frac{15D^3}{8}$.

step 4 $D(x^3 + x^2) = 3x^2 + 2x$, $D^2 = 6x + 2$, $D^3 = 6$: $\frac{1}{2}\left[x^3 + x^2 - \frac{9x^2+6x}{2} + \frac{7(6x+2)}{4} - \frac{90}{8}\right]$.

answer

$$y = c_1 e^{-x} + c_2 e^{-2x} + \frac{1}{2}\left[x^3 - \frac{7x^2}{2} + \frac{30x}{4} - \frac{31}{4}\right]$$

check Slide answer agrees; sympy dsolve gives $\frac{x^3}{2} - \frac{7x^2}{4} + \frac{15x}{4} - \frac{31}{8}$, the same. Figure G1.1.

S005·11 · S005·12 · T006a 00:02:37 · T006a 00:07:23

W2.2 · $(2D^2+2D+3)y = x^2+2x-1$ (slides 005·13–14) ESA ★★★

GIVEN Solve $(2D^2+2D+3)y = x^2+2x-1$.

STEP 1 AE $2m^2+2m+3=0$: $m = \frac{-2\pm\sqrt{-20}}{4} = \frac{-1\pm\sqrt{5}i}{2}$ (slide prints $\frac{-2\pm\sqrt{5}i}{2}$).

STEP 2 CF $= e^{-x/2}\left(c_1\cos\frac{\sqrt{5}}{2}x + c_2\sin\frac{\sqrt{5}}{2}x\right)$ (slide drops c_2).

STEP 3 PI $= \frac{1}{3}\left[1+\frac{2D^2+2D}{3}\right]^{-1}(x^2+2x-1)$; up to D^2: $1-\frac{2D^2+2D}{3}+\frac{4D^2}{9} = 1-\frac{2D}{3}-\frac{2D^2}{9}$.

STEP 4 $D(x^2+2x-1) = 2x+2$, $D^2 = 2$: $\frac{1}{3}\left[x^2+2x-1-\frac{4x+4}{3}-\frac{4}{9}\right]$.

ANSWER

$$\boxed{y = e^{-x/2}\left(c_1\cos\frac{\sqrt{5}}{2}x + c_2\sin\frac{\sqrt{5}}{2}x\right) + \frac{1}{3}\left(x^2+\frac{2x}{3}-\frac{25}{9}\right)}$$

CHECK PI agrees with the slide; sympy dsolve and checkodesol agree (roots $-\frac{1}{2}\pm\frac{\sqrt{5}}{2}i$).

S005·13 · S005·14 · T006a 00:07:45 · T006a 00:09:22

W2.3 · $y''' - y = x^5+3x^4-2x^3$ (CW P16) ESA ★★

GIVEN Solve $y''' - y = x^5+3x^4-2x^3$, i.e. $(D^3-1)y = X$.

STEP 1 AE $(m-1)(m^2+m+1)=0$: $m=1,\ -\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$. CF $= c_1e^x + e^{-x/2}\left(c_2\cos\frac{\sqrt{3}}{2}x + c_3\sin\frac{\sqrt{3}}{2}x\right)$.

STEP 2 PI $= -(1-D^3)^{-1}X = -(1+D^3+D^6+\dots)X$; $m=5$, so keep $1+D^3$.

STEP 3 $D^3X = 60x^2+72x-12$ (from $X' = 5x^4+12x^3-6x^2$, $X'' = 20x^3+36x^2-12x$).

ANSWER

$$\boxed{y = c_1e^x + e^{-x/2}\left(c_2\cos\frac{\sqrt{3}}{2}x + c_3\sin\frac{\sqrt{3}}{2}x\right) - x^5-3x^4+2x^3-60x^2-72x+12}$$

CHECK CW answer agrees; sympy dsolve and checkodesol agree.

CW U2 P16

SHORT-NOTE RECAP

$(1+z)^{-1} = 1-z+z^2-z^3$, $(1-z)^{-1} = 1+z+z^2+z^3$, $(1+z)^{-2} = 1-2z+3z^2-4z^3$, $(1-z)^{-2} = 1+2z+3z^2+4z^3$. In squares and cubes keep only products with D-power $\le m$: for $m=3$, $(D^2+3D)^2 \to 9D^2+6D^3$, $(D^2+3D)^3 \to 27D^3$. $\frac{x^3+x^2}{D^2+3D+2} = \frac{1}{2}(x^3-\frac{7x^2}{2}+\frac{15x}{2}-\frac{31}{4})$; $\frac{x^2+2x-1}{2D^2+2D+3} = \frac{1}{3}(x^2+\frac{2x}{3}-\frac{25}{9})$; CW P16: $\frac{X}{D^3-1} = -(1+D^3)X$.

3 When D is a factor of $f(D)$

If $f(D)$ has no constant term, the lowest-degree term is D (or D^k), and taking it out leaves $\frac{1}{D^k}$ in front of the series. $\frac{1}{D}$ is integration without a constant, so the PI comes out one degree higher than X for each factor D. The slides show no such example; it is added for clarity because it follows directly from the slide rule.

Key terms and definitions

$\frac{1}{D}$. The inverse of D: integration with respect to x, without a constant of integration (class 003).
ADDED FOR CLARITY S005·4

Formulas

E3.1 · D as a factor ESA ★

$$\frac{1}{D^k\,\psi(D)}x^m = \frac{1}{D^k}\Big[\big(\psi(D)\big)^{-1}x^m\Big], \qquad \frac{1}{D} = \int (\,)\,dx$$

GIVES The PI when $f(D)$ has no constant term.
APPLIES $f(D) = D^k\psi(D)$, $\psi(0) \neq 0$.
NOTE Expand $\psi(D)^{-1}$ up to D^m, operate, then integrate k times (no constants). The PI has degree $m + k$.
ADDED FOR CLARITY S005·4

E3.2 · A constant right-hand side ESA ★★

$$\frac{1}{f(D)}k = \frac{k}{f(0)}\ (f(0) \neq 0), \qquad \frac{1}{D\,\psi(D)}k = \frac{kx}{\psi(0)}$$

GIVES The PI of the constant term of a polynomial or of $\sin^2$, $\cos^2$.
APPLIES $m = 0$: $k = kx^0 = ke^{0x}$.
NOTE Type 1 with $a = 0$ (class 003) and Type 3 with $m = 0$ agree.
ADDED FOR CLARITY ESA Jan-24 s4 2c

Graphs and diagrams

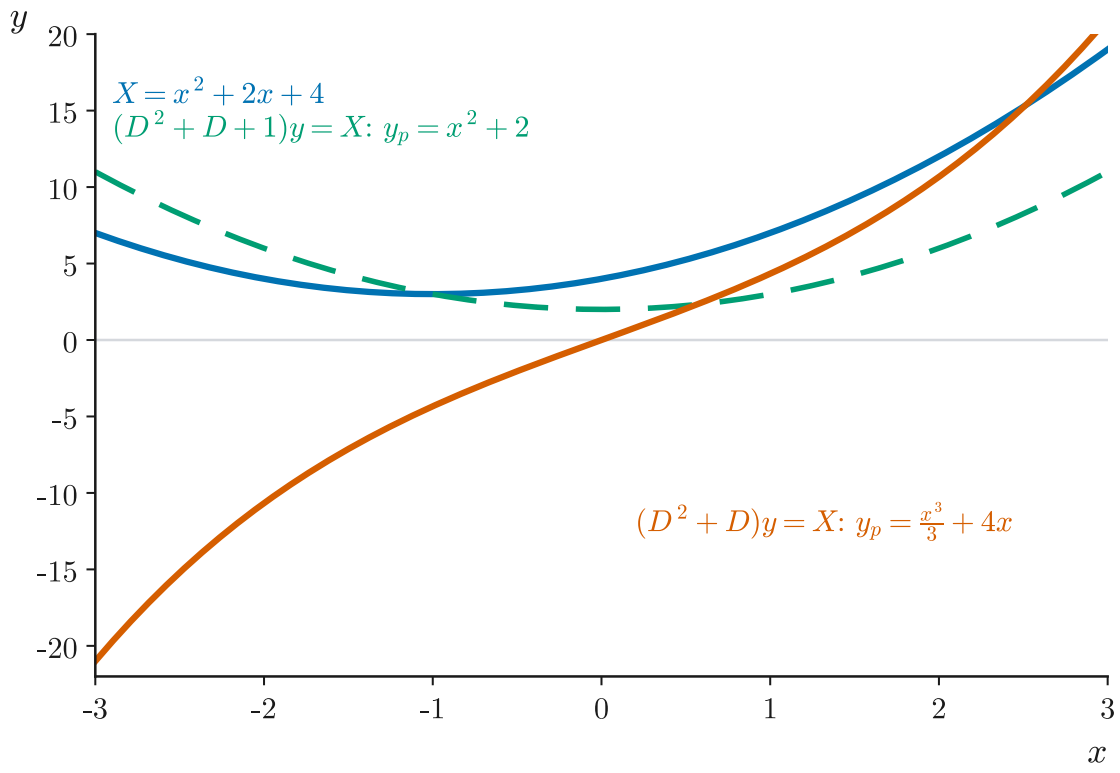
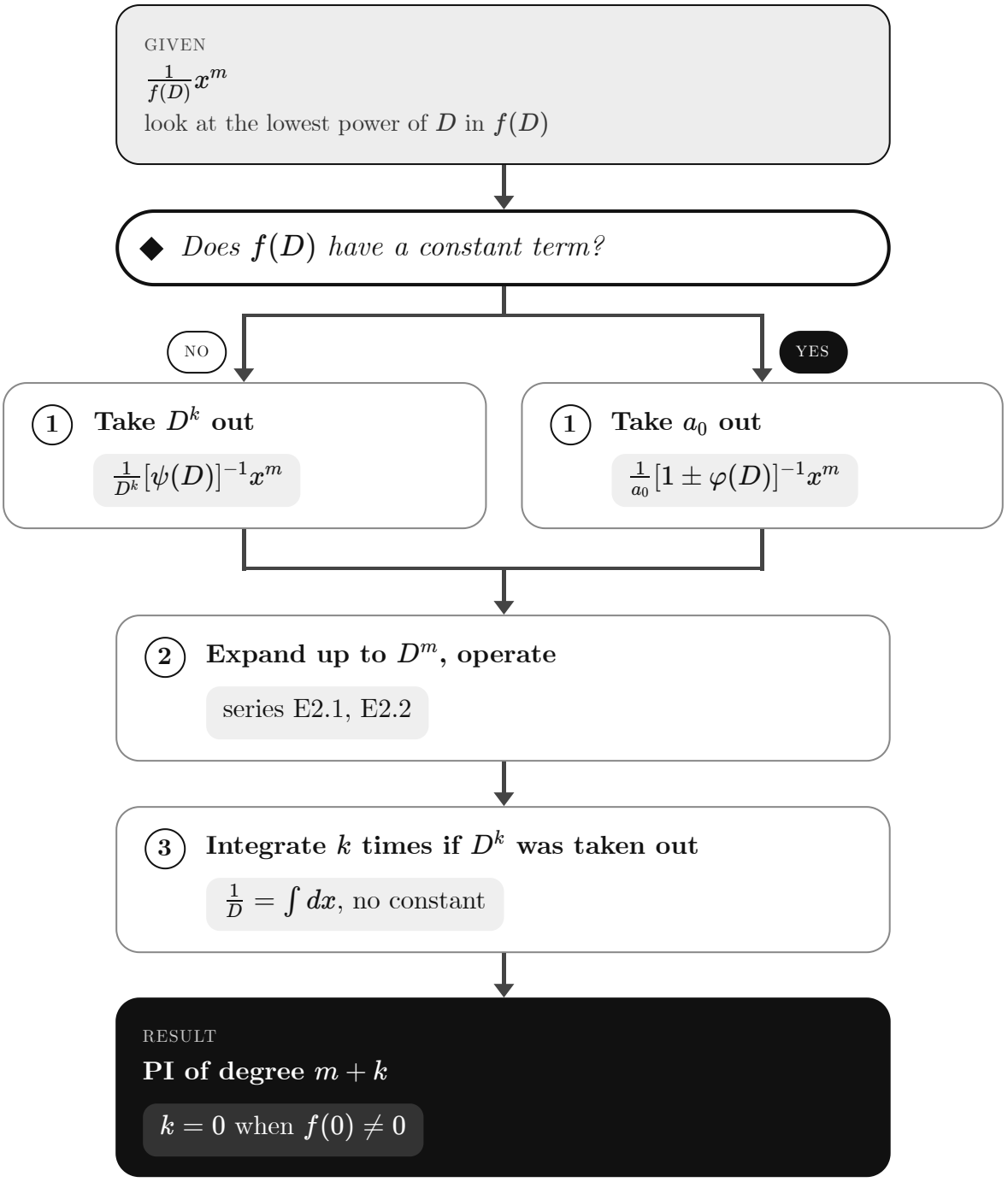


Figure G3.1. A factor D raises the degree of the PI. ESA ★

ADDED FOR CLARITY S005·4

How to read it. Same right-hand side $X = x^2 + 2x + 4$ (blue). With $f(D) = D^2 + D + 1$ ($f(0) = 1$) the PI $x^2 + 2$ (green dashed) is a quadratic like X. With $f(D) = D^2 + D$ (D a factor) the PI $\frac{x^3}{3} + 4x$ (vermilion) is a cubic: the $\frac{1}{D}$ integrates once.

Flowcharts



Remark. $f(0) = 0$ means $m = 0$ is an AE root: the CF already contains a constant.

Figure F3.1. Constant term or factor D? ESA ★

ADDED FOR CLARITY S005·4

Important points and why

1. If $f(D)$ has no constant term, its lowest-degree term is D^k: take it out, expand the rest, operate, then integrate k times. The PI has degree $m + k$. ESA ★ ADDED FOR CLARITY S005·4

2. When integrating ($\frac{1}{D}$) add no constant: constants belong to the CF, which already has c_1 (the root $m = 0$). ESA ★ ADDED FOR CLARITY S005·4

Why does a factor D raise the degree of the PI?
If $f(D) = D\psi(D)$, then $f(D)$ applied to a polynomial of degree m gives degree $m - 1$ at most, so to reproduce x^m the PI must have degree $m + 1$. The constant part of the PI is then a CF term ($m = 0$ is an AE root) and is dropped.

ADDED FOR CLARITY S005·4

Exam trap. With a factor D, do not add constants when integrating, and do not stop the expansion early: integrate after operating. ADDED FOR CLARITY S005·4

Worked examples

W3.1 · $(D^2 + D)y = x^2 + 2x + 4$: a factor D ESA ★

GIVEN Solve $(D^2 + D)y = x^2 + 2x + 4$.

STEP 1 AE $m(m + 1) = 0$: CF $= c_1 + c_2 e^{-x}$.

STEP 2 Lowest-degree term D: PI $= \frac{1}{D}(1 + D)^{-1}(x^2 + 2x + 4) = \frac{1}{D}(1 - D + D^2)(x^2 + 2x + 4)$.

STEP 3 $(1 - D + D^2)(x^2 + 2x + 4) = x^2 + 2x + 4 - (2x + 2) + 2 = x^2 + 4$.

STEP 4 $\frac{1}{D}(x^2 + 4) = \int (x^2 + 4)\,dx = \frac{x^3}{3} + 4x$ (no constant).

ANSWER $\boxed{y = c_1 + c_2 e^{-x} + \frac{x^3}{3} + 4x}$

CHECK sympy dsolve and checkodesol agree. Figure G3.1 compares $(D^2 + D + 1)y =$ the same X, whose PI $x^2 + 2$ keeps degree 2.

ADDED FOR CLARITY S005·4

W3.2 · $(D^3 + D^2)y = 6x$: a factor D^2 ESA ★

GIVEN Solve $(D^3 + D^2)y = 6x$.

STEP 1 AE $m^2(m + 1) = 0$: CF $= c_1 + c_2 x + c_3 e^{-x}$.

STEP 2 PI $= \frac{1}{D^2}(1 + D)^{-1}6x = \frac{1}{D^2}(1 - D)6x = \frac{1}{D^2}(6x - 6)$.

STEP 3 Integrate twice: $\frac{1}{D}(6x - 6) = 3x^2 - 6x$, $\frac{1}{D}(3x^2 - 6x) = x^3 - 3x^2$.

ANSWER $\boxed{y = c_1 + c_2 x + c_3 e^{-x} + x^3 - 3x^2}$

CHECK sympy dsolve and checkodesol agree.

ADDED FOR CLARITY S005·4

SHORT-NOTE RECAP

No constant term in $f(D)$: take D^k out, $\frac{1}{D^k}[\psi(D)]^{-1}x^m$; expand, operate, then integrate k times without constants. The PI rises k degrees. $\frac{x^2+2x+4}{D^2+D} = \frac{x^3}{3} + 4x$; $\frac{6x}{D^3+D^2} = x^3 - 3x^2$. A constant: $\frac{k}{f(D)} = \frac{k}{f(0)}$.

4 Mixed right-hand sides: Types 1, 2 and 3 together

Every ESA question with a polynomial term also has e^{ax}, a^x, $\sin ax$ or $\cos ax$ in X. The operator is linear, so the PI of a sum is the sum of the PIs: split X, give each term its own rule (Type 1 for e^{ax} and $a^x = e^{x \log a}$, Type 2 for $\sin$ and $\cos$, Type 3 for the polynomial, with the cases of failure where the CF demands them) and add. This is the most-asked PI question in the eight papers: 4 distinct questions, all solved here.

Key terms and definitions

Mixed right-hand side. X is a sum of terms of different types, e.g. $e^{4x} + \sin x + x^2$. ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c

Formulas

E4.1 · Superposition ESA ★★★

$$\frac{1}{f(D)}(X_1 + X_2 + X_3) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 + \frac{1}{f(D)}X_3 = \mathrm{PI}_1 + \mathrm{PI}_2 + \mathrm{PI}_3$$

GIVES The PI of a mixed right-hand side.

APPLIES X is a sum of terms of different types.

NOTE Each term chooses its own rule and its own failure test.

ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c · CW U2 P21

E4.2 · Reminder: e^{ax} and $a^{\pm x}$ (Type 1) ESA ★★★

$$\frac{e^{ax}}{f(D)} = \frac{e^{ax}}{f(a)}, \qquad \frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)} \qquad (f \neq 0)$$

GIVES PI of the exponential terms; on failure $x\frac{e^{ax}}{f'(a)}$, then $x^2\frac{e^{ax}}{f''(a)}$.

APPLIES $a^{-x} = e^{-x \log a}$, so $D \to -\log a$.

NOTE Taught in the class 003 pack; 2^{-x} is in two ESA questions.

ESA Jan-24 s1 2d · ESA Jan-24 s4 2c · ESA Jan-24 s2 2d

E4.3 · Reminder: $\sin ax$, $\cos ax$ (Type 2) ESA ★★★

$$\frac{\sin ax}{f(D^2)} = \frac{\sin ax}{f(-a^2)}, \qquad \frac{\cos ax}{D^2 + a^2} = \frac{x \sin ax}{2a}$$

GIVES PI of the trigonometric terms, with the failure case.

APPLIES Replace D^2 by $-a^2$; rationalise a lone D.

NOTE Taught in the class 004 pack.

ESA Jul-23 3b · ESA Jan-24 s1 2d · CW U2 P21

Graphs and diagrams

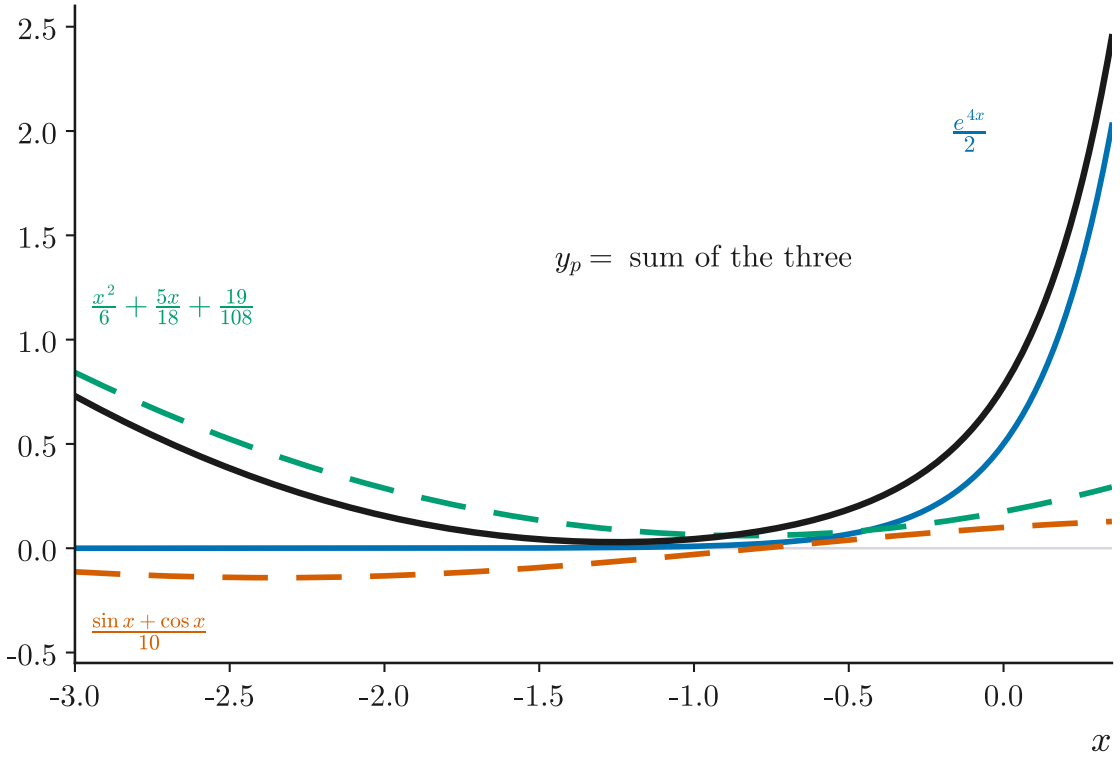
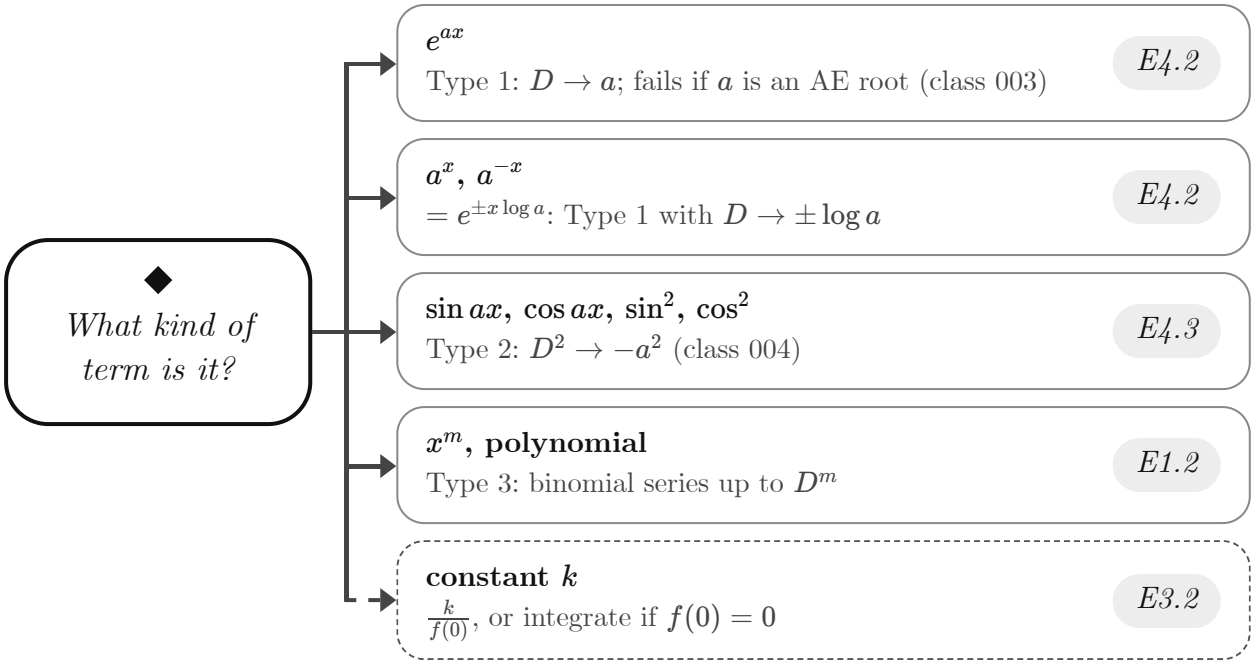


Figure G4.1. Superposition in ESA Jul-23 3b: three PIs added. ESA ★★★

ADDED FOR CLARITY ESA Jul-23 3b

How to read it. The three parts of the PI of $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$: $\frac{e^{4x}}{2}$ (blue, Type 1), $\frac{\sin x + \cos x}{10}$ (vermilion dashed, Type 2), $\frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$ (green dashed, Type 3), and their sum (black). Each part is found separately; the PI is the plain sum.

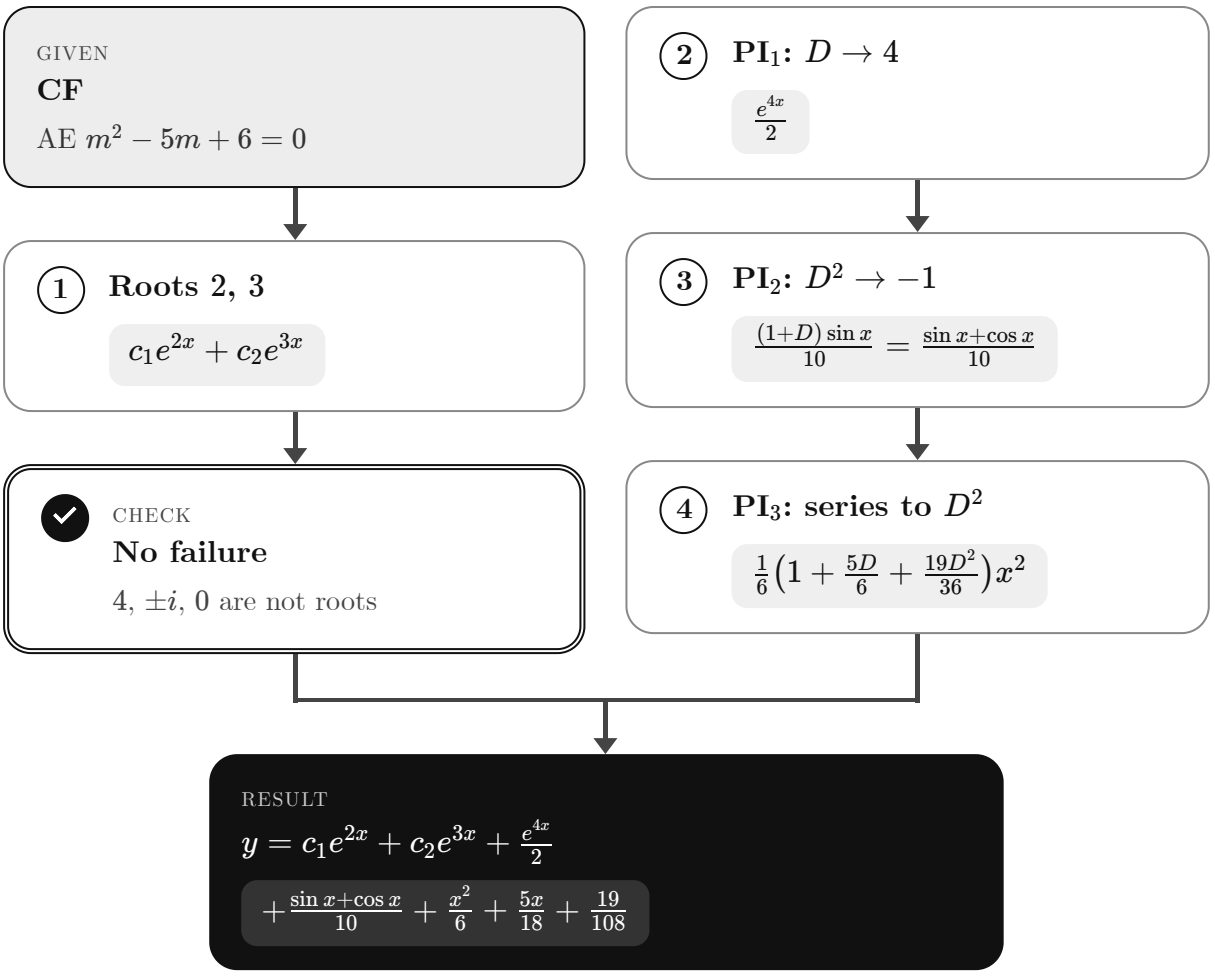
Flowcharts



Remark. Then add: PI = PI_1+PI_2 + … (E4.1).

Figure F4.1. Give each term of X its rule ESA ★★★

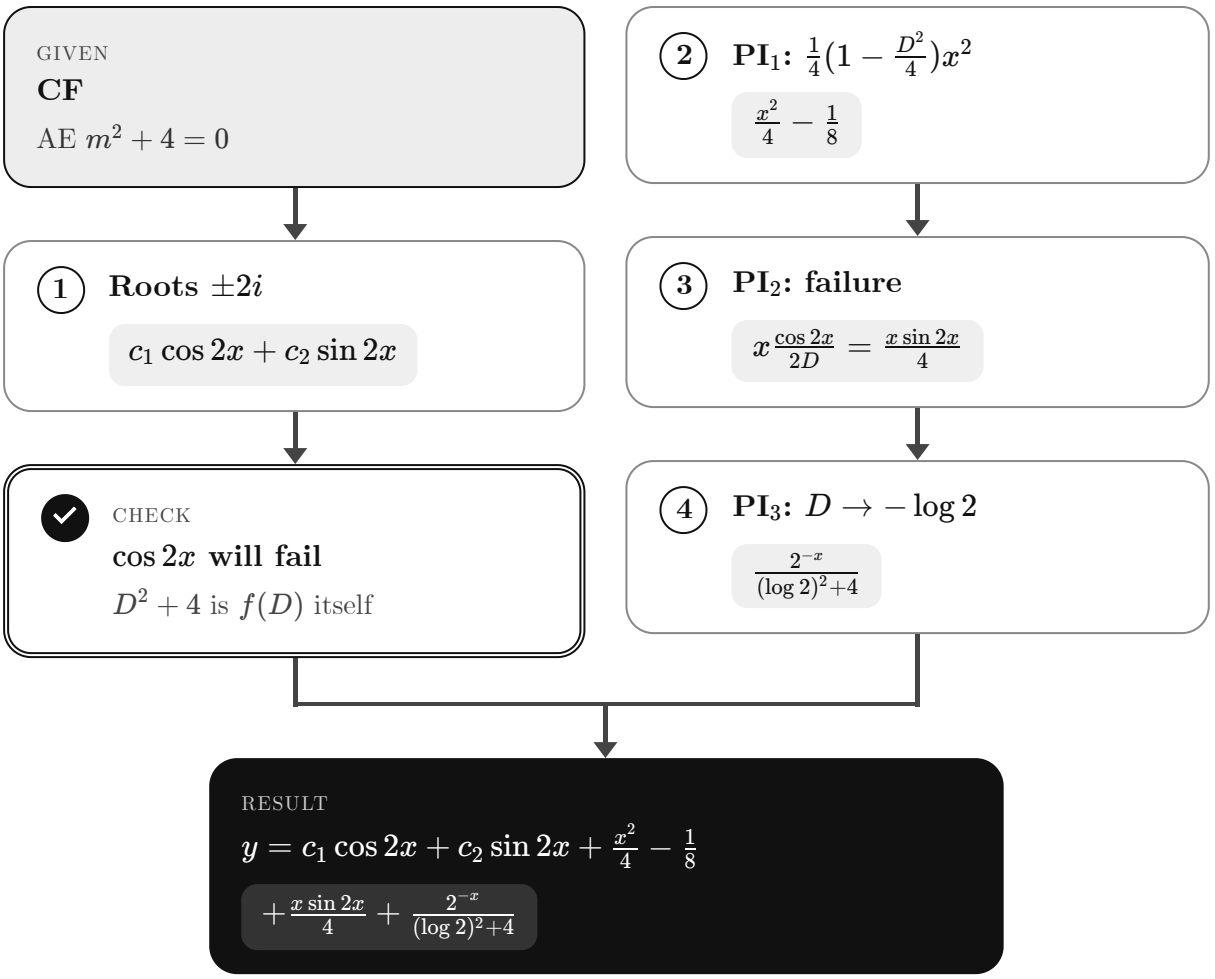
ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c



Remark. Worked in full as W4.1 (7 marks).

Figure F4.2. ESA Jul-23 3b: $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$ ESA ★★★

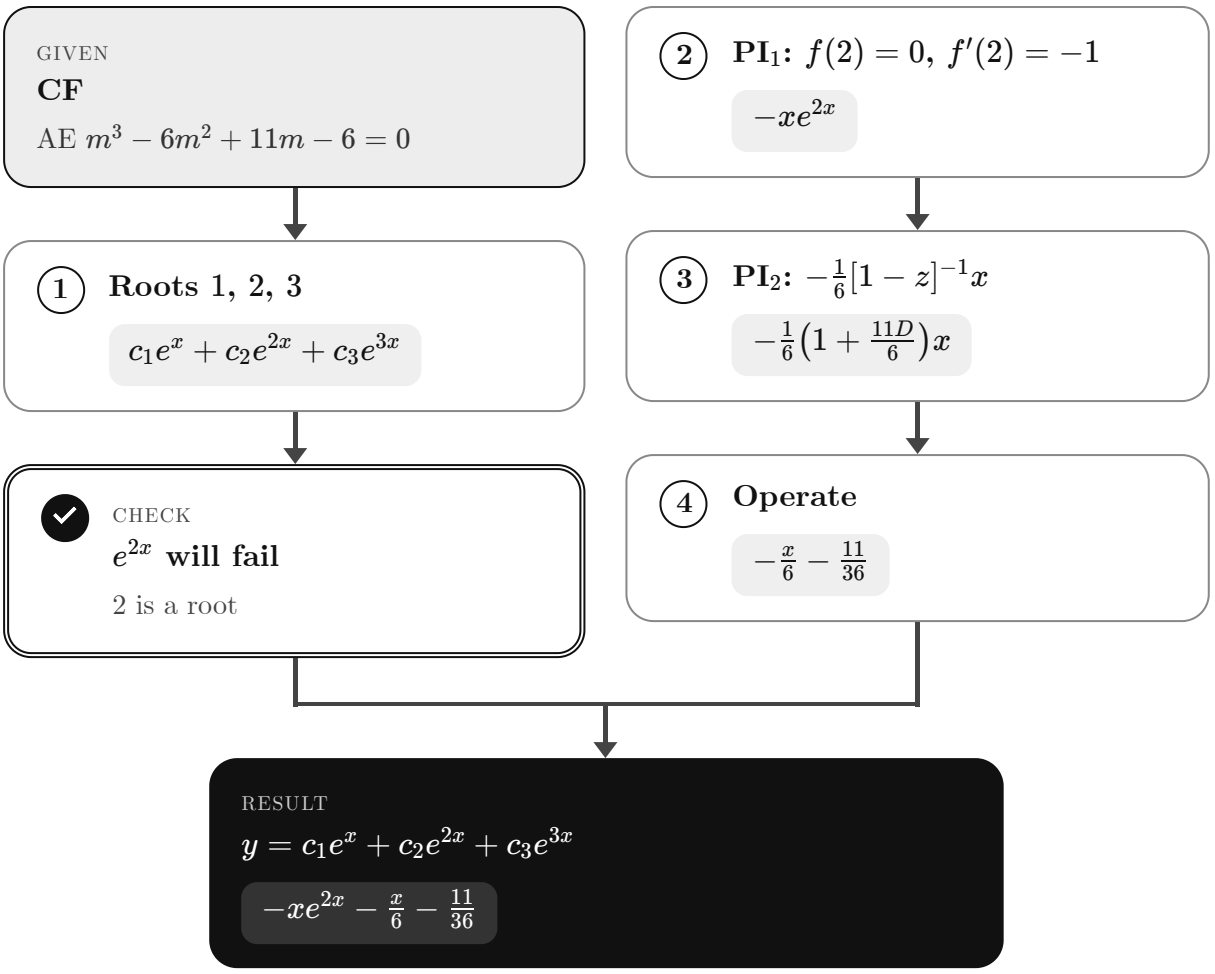
ESA Jul-23 3b



Remark. Worked in full as W4.2 (5 marks).

Figure F4.3. ESA Jan-24 s1 2d: $(D^2 + 4)y = x^2 + \cos 2x + 2^{-x}$ ESA ★★★

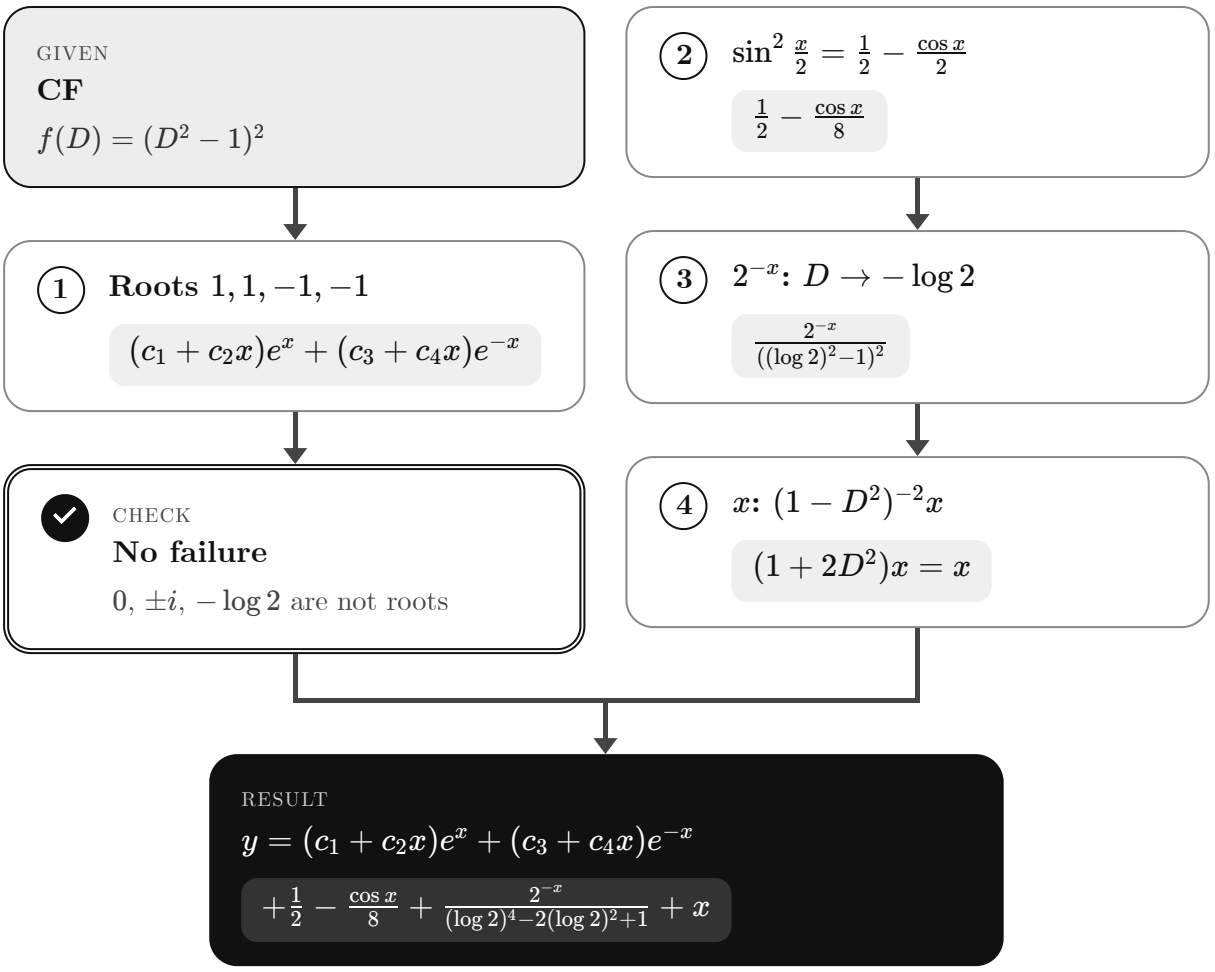
ESA Jan-24 s1 2d



Remark. For $X = x$ ($m = 1$) only $1 + z$ of the series is needed, and of z only its D term. Worked as W4.3.

Figure F4.4. ESA Jan-24 s2 2d: $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$ ESA ★★★

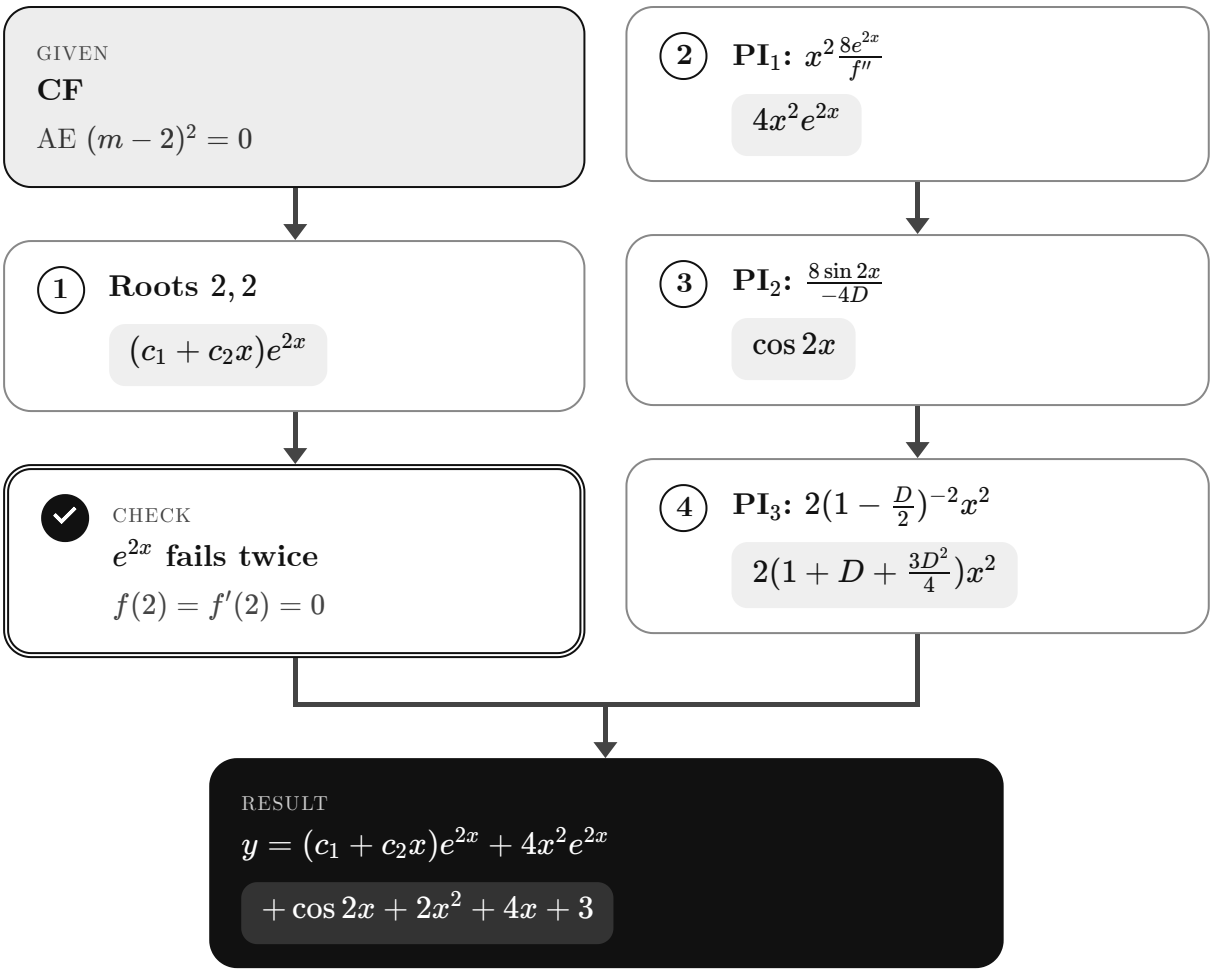
ESA Jan-24 s2 2d



Remark. Four terms, three rules. Worked in full as W4.4 (6 marks).

Figure F4.5. ESA Jan-24 s4 2c: $(D-1)^2(D+1)^2y = \sin^2 \frac{x}{2} + 2^{-x} + x$
ESA ★★★

ESA Jan-24 s4 2c · CW U2 P22



Remark. The ESA question Dec-23 3a (asked three times) is this without x^2. Worked as W4.5.

Figure F4.6. CW P21: $(D-2)^2y=8(e^{2x}+\sin 2x+x^2)$ ESA ★★★

CW U2 P21 · ESA Dec-23 3a

Every mixed question with a polynomial term

Paper, Q	$f(D)$	Other terms (rule)	Polynomial term → PI	Worked
Jul-23 3b	$D^2 - 5D + 6$	e^{4x} (T1), $\sin x$ (T2)	$x^2 \to \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$	W4.1
Jan-24 s1 2d	$D^2 + 4$	$\cos 2x$ (T2, fails), 2^{-x} (T1)	$x^2 \to \frac{x^2}{4} - \frac{1}{8}$	W4.2
Jan-24 s2 2d	$D^3 - 6D^2 + 11D - 6$	e^{2x} (T1, fails)	$x \to -\frac{x}{6} - \frac{11}{36}$	W4.3
Jan-24 s4 2c (= CW P22)	$(D-1)^2(D+1)^2$	$\sin^2 \frac{x}{2}$ (T2 + const), 2^{-x} (T1)	$x \to x$	W4.4
CW P21	$(D-2)^2$	$8e^{2x}$ (T1, fails twice), $8\sin 2x$ (T2)	$8x^2 \to 2x^2 + 4x + 3$	W4.5

ADDED FOR CLARITY ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c · CW U2 P21

Important points and why

1. Split X into its terms first, label PI_1, PI_2, PI_3, and add at the end (E4.1). ESA ★★★ ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c
2. Find the CF first: its roots tell you which terms fail (a a root: e^{ax} fails; $\pm ai$ roots: $\sin ax$, $\cos ax$ fail; 0 a root: D is a factor for the polynomial). ESA ★★★ ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · CW U2 P21
3. $a^{-x} = e^{-x \log a}$, so 2^{-x} is Type 1 with $D \to -\log 2$: $(D^2 + 4) \to (\log 2)^2 + 4$. ESA ★★★ ESA Jan-24 s1 2d · ESA Jan-24 s4 2c
4. $\sin^2 \frac{x}{2} = \frac{1 - \cos x}{2}$ gives a constant (E3.2) and a Type 2 term; the polynomial x is a separate Type 3 term. ESA ★★★ ESA Jan-24 s4 2c · CW U2 P22

Exam trap. Never put x^2 into a Type 1 or Type 2 formula, and never expand in powers of D for e^{ax}: each term keeps its own rule. ADDED FOR CLARITY ESA Jul-23 3b

Worked examples

W4.1 · $(D^2-5D+6)y = e^{4x}+\sin x + x^2$ (ESA) — ESA ★★★

given Solve $(D^2-5D+6)y = e^{4x}+\sin x + x^2$.

step 1 AE $(m-2)(m-3)=0$: CF $= c_1e^{2x}+c_2e^{3x}$. No term fails (4, $\pm i$, 0 are not roots).

step 2 $\text{PI}_1 = \frac{e^{4x}}{16-20+6} = \frac{e^{4x}}{2}$ (Type 1).

step 3 PI_2: $D^2\to -1$: $\frac{\sin x}{5-5D} = \frac{(1+D)\sin x}{5(1-D^2)} = \frac{(1+D)\sin x}{10} = \frac{\sin x+\cos x}{10}$ (Type 2).

step 4 $\text{PI}_3 = \frac{1}{6}\left[1+\frac{D^2-5D}{6}\right]^{-1}x^2$; up to D^2: $1-\frac{D^2-5D}{6}+\frac{25D^2}{36} = 1+\frac{5D}{6}+\frac{19D^2}{36}$.

step 5 $\text{PI}_3 = \frac{1}{6}\left[x^2+\frac{5x}{3}+\frac{19}{18}\right] = \frac{x^2}{6}+\frac{5x}{18}+\frac{19}{108}$.

answer $$\boxed{y = c_1e^{2x}+c_2e^{3x}+\frac{e^{4x}}{2}+\frac{\sin x+\cos x}{10}+\frac{x^2}{6}+\frac{5x}{18}+\frac{19}{108}}$$

check sympy dsolve and checkodesol agree. Figure G4.1 plots the three parts.

ESA Jul-23 3b

W4.2 · $(D^2+4)y = x^2+\cos 2x + 2^{-x}$ (ESA) — ESA ★★★

given Solve $(D^2+4)y = x^2+\cos 2x+2^{-x}$.

step 1 AE $m^2+4=0$: $m=\pm 2i$. CF $= c_1\cos 2x + c_2\sin 2x$. So $\cos 2x$ will fail.

step 2 $\text{PI}_1 = \frac{1}{4}\left[1+\frac{D^2}{4}\right]^{-1}x^2 = \frac{1}{4}\left[1-\frac{D^2}{4}\right]x^2 = \frac{1}{4}\left(x^2-\frac{1}{2}\right) = \frac{x^2}{4}-\frac{1}{8}$.

step 3 PI_2: $D^2\to -4$ gives 0, failure: $x\frac{\cos 2x}{2D} = \frac{x}{2}\int\cos 2x\,dx = \frac{x\sin 2x}{4}$.

step 4 PI_3: $2^{-x} = e^{-x\log 2}$, $D\to -\log 2$: $\frac{2^{-x}}{(\log 2)^2+4}$.

answer $$\boxed{y = c_1\cos 2x + c_2\sin 2x + \frac{x^2}{4}-\frac{1}{8}+\frac{x\sin 2x}{4}+\frac{2^{-x}}{(\log 2)^2+4}}$$

check sympy dsolve and checkodesol agree.

ESA Jan-24 s1 2d

W4.3 · $(D^3-6D^2+11D-6)y=e^{2x}+x$ (ESA) ESA ★★★

GIVEN Solve $(D^3-6D^2+11D-6)y=e^{2x}+x$.

STEP 1 AE $(m-1)(m-2)(m-3)=0$: CF $=c_1e^x+c_2e^{2x}+c_3e^{3x}$. So e^{2x} will fail.

STEP 2 PI_1: $f(2)=8-24+22-6=0$; $f'(D)=3D^2-12D+11$, $f'(2)=-1$: $\text{PI}_1=x\frac{e^{2x}}{-1}=-xe^{2x}$.

STEP 3 $\text{PI}_2=-\frac{1}{6}\left[1-\frac{11D-6D^2+D^3}{6}\right]^{-1}x$; $m=1$, so $(1-z)^{-1}\to 1+z\to 1+\frac{11D}{6}$.

STEP 4 $\text{PI}_2=-\frac{1}{6}\left(x+\frac{11}{6}\right)=-\frac{x}{6}-\frac{11}{36}$.

ANSWER $$\boxed{y=c_1e^x+c_2e^{2x}+c_3e^{3x}-xe^{2x}-\frac{x}{6}-\frac{11}{36}}$$

CHECK sympy dsolve and checkodesol agree.

ESA Jan-24 s2 2d

W4.4 · $(D-1)^2(D+1)^2y=\sin^2\frac{x}{2}+2^{-x}+x$ (ESA; CW P22) ESA ★★★

GIVEN Solve $[(D-1)^2(D+1)^2]y=\sin^2\frac{x}{2}+2^{-x}+x$.

STEP 1 $f(D)=(D^2-1)^2$; AE roots $1,1,-1,-1$: CF $=(c_1+c_2x)e^x+(c_3+c_4x)e^{-x}$. No term fails.

STEP 2 $\sin^2\frac{x}{2}=\frac{1}{2}-\frac{\cos x}{2}$. Constant: $\frac{1}{2f(0)}=\frac{1}{2}$. Cosine: $D^2\to -1$: $(-1-1)^2=4$: $-\frac{\cos x}{8}$.

STEP 3 2^{-x}: $D\to-\log 2$: $f=\left((\log 2)^2-1\right)^2=(\log 2)^4-2(\log 2)^2+1$.

STEP 4 x: $\frac{1}{(1-D^2)^2}x=(1-D^2)^{-2}x=(1+2D^2+\dots)x=x$ (E2.2, $m=1$).

ANSWER $$\boxed{y=(c_1+c_2x)e^x+(c_3+c_4x)e^{-x}+\frac{1}{2}-\frac{\cos x}{8}+\frac{2^{-x}}{(\log 2)^4-2(\log 2)^2+1}+x}$$

CHECK CW answer agrees; sympy dsolve and checkodesol agree.

ESA Jan-24 s4 2c · CW U2 P22

W4.5 · $(D-2)^2y = 8(e^{2x} + \sin 2x + x^2)$ (CW P21) ESA ★★★

GIVEN Solve $(D-2)^2y = 8(e^{2x} + \sin 2x + x^2)$ (the ESA question Dec-23 3a plus the term $8x^2$).

STEP 1 AE $(m-2)^2 = 0$: CF $= (c_1 + c_2x)e^{2x}$. e^{2x} fails twice.

STEP 2 PI_1: $f(2) = f'(2) = 0$, $f'' = 2$: $8 \cdot \frac{x^2e^{2x}}{2} = 4x^2e^{2x}$. PI_2: $D^2 \to -4$: $\frac{8\sin 2x}{-4D} = -2\int \sin 2x\,dx = \cos 2x$.

STEP 3 $\text{PI}_3 = \frac{8x^2}{4(1-\frac{D}{2})^2} = 2\big(1-\frac{D}{2}\big)^{-2}x^2 = 2\big(1 + D + \frac{3D^2}{4}\big)x^2$ (E2.2 with $z = \frac{D}{2}$).

STEP 4 $\text{PI}_3 = 2\big(x^2 + 2x + \frac{3}{2}\big) = 2x^2 + 4x + 3$.

ANSWER $\boxed{y = (c_1 + c_2x)e^{2x} + 4x^2e^{2x} + \cos 2x + 2x^2 + 4x + 3}$

CHECK CW answer agrees; sympy dsolve and checkodesol agree. Without x^2 this is ESA Dec-23 3a (class 004 pack).

CW U2 P21 · ESA Dec-23 3a

Past-paper questions

ESA Jul 2023 · 7 marks · UE22MA141A Q3.b · solved in W4.1

Solve $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$.

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q2.d · solved in W4.2

Solve $(D^2 + 4)y = x^2 + \cos 2x + 2^{-x}$.

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q2.d · solved in W4.3

Solve $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (= CW P22) · solved in W4.4

Solve $[(D-1)^2(D+1)^2]y = \sin^2\frac{x}{2} + 2^{-x} + x$.

SHORT-NOTE RECAP

Mixed X: CF first (it shows which terms fail), then PI_1+PI_2+PI_3, each with its own rule: $e^{ax} \to D = a$; $a^{-x} = e^{-x\log a} \to D = -\log a$; $\sin, \cos \to D^2 = -a^2$; polynomial → series. ESA: $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$: PI $= \frac{e^{4x}}{2} + \frac{\sin x + \cos x}{10} + \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$. $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$: PI $= -xe^{2x} - \frac{x}{6} - \frac{11}{36}$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant	E1.3, E3.1
$f(D)$	Operator polynomial	Left-hand side operator of $f(D)y = X$	E1.1
a_0	Lowest-degree term	Taken out of $f(D)$ with its sign (usually the constant term)	E1.2
$\varphi(D)$	Phi of D	Rest of $f(D)$ after a_0 is taken out; only positive powers of D	E1.2
$\psi(D)$	Psi of D	$f(D)/D^k$ when D^k is a factor; $\psi(0) \neq 0$	E3.1
z	Series variable	Stands for $\varphi(D)$; slide 005·5 writes it as x (not the variable x)	E2.1, E2.2
m	Degree of X	Expand up to D^m; slide 005·4 calls it n. Also the AE unknown (same letter)	E1.1, E1.3
k	Power of the factor D	Number of integrations; also a constant in E3.2	E3.1, E3.2
PI_1, PI_2, PI_3	Partial PIs	PIs of the separate terms of X	E4.1
X	Right-hand side	A function of x; capital X, small x	E1.1

B Sources, coverage and gaps

Source	Path	Use
Deck 005 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/005_Particular_integral_of_Type_3__50cbfe7011ebf0b0.*	15 slides; content on 3–14 (every equation read from the rendered images)
Transcript T005 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.*	12.6 min; slides 4–9
Transcript T006a (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/006_Particular_integral_-Type_3_and_Type_4__fb358573adb09195.*	15.5 min; 00:00:00–00:09:55 Type 3 and slides 10–14 (the rest is Type 4, used by the class 006 pack)
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problems 15, 16, 21 and 22 (all solved in full; all four printed answers are right)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Jul-23 3b, Jan-24 set 1 2d, set 2 2d, set 4 2c, via the shared ESA map
Video 005 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/005_Particular_integral_of_standard_functions_-_Type_3__1f8ea5ed4d7e088f.mp4	Not opened: every slide equation is readable and no video trigger applies

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences; 7 distinct / 9 occurrences are PI by operator rules). A Type 3 term sits in 4 distinct questions (4 occurrences), every one of them a mixed right-hand side: Jul-23 3b $(D^2-5D+6)y=e^{4x}+\sin x+x^2$ (7 marks), Jan-24 set 1 2d $(D^2+4)y=x^2+\cos 2x+2^{-x}$ (5), set 2 2d $(D^3-6D^2+11D-6)y=e^{2x}+x$ (5) and set 4 2c $(D-1)^2(D+1)^2y=\sin^2\frac{x}{2}+2^{-x}+x$ (6, = CW P22). All four are solved in full here, with CW P15, P16 and P21. Stars: the Type 3 rule, the expansions and superposition ★★★ (4 questions); the case $f(D)$ with a factor D ★ (never asked); the second-order expansion $(1\pm z)^{-2}$ ★★ (CW P21 and P22 need it, no paper asks it on its own).

Known gaps, conflicts and corrections

1. Slide 005·7 expands $-(1-D^2)^{-1}$ as $-(1-D^2+D^4-\dots)$; it is $-(1+D^2+D^4+\dots)$, because $(1-z)^{-1}$ has only plus signs (slide 005·5). The lecturer corrects it aloud (T005 00:07:07). The answer $2-5x$ is unaffected, since $D^2(5x-2)=0$.
2. Slide 005·13 gives the roots of $2m^2+2m+3=0$ as $\frac{-2\pm\sqrt{5}i}{2}$; they are $\frac{-1\pm\sqrt{5}i}{2}$ $(=\frac{-2\pm\sqrt{-20}}{4})$. The CF it writes, $e^{-x/2}(\dots\frac{\sqrt{5}}{2}x)$, uses the correct roots, and so does the lecture (T006a 00:07:45).
3. Slides 005·13 and 005·14 write the CF as $e^{-x/2}\big(c_1\cos\frac{\sqrt{5}}{2}x+\sin\frac{\sqrt{5}}{2}x\big)$: the constant c_2 is missing on the sine term. A second-order equation needs two arbitrary constants.

4. Slide 005·3 reads "$f(D)y = x$ when $x = x^n$"; it means $f(D)y = X$ with $X = x^n$ (capital X).
5. T006a 00:02:23 says "the nth derivative of x^m is zero when n is m"; it is zero when $n > m$ (the mth derivative is the constant $m!$), as the lecture itself said at 00:01:29 and slide 005·4 says.
6. The slides never show $f(D)$ without a constant term; the case of a factor D (Topic 3, W3.1, W3.2) is added for clarity. It follows the slide rule literally: the lowest-degree term is then D (or D^2), and $\frac{1}{D}$ is integration (class 003 pack).
7. Slides 005·8–9 ask only for the PI; slides 005·6–7 stop at "the GS is $y = y_c + y_p$": the general solutions are written out here.
8. All slide answers were recomputed with sympy (dsolve, checkodesol): 005·7 $2 - 5x$, 005·9 $\frac{1}{4}(x^2 + \frac{9}{2}x + \frac{23}{8})$, 005·12 $\frac{1}{2}(x^3 - \frac{7x^2}{2} + \frac{30x}{4} - \frac{31}{4})$ and 005·14 $\frac{1}{3}(x^2 + \frac{2x}{3} - \frac{25}{9})$ are right. The CW answers of P15, P16, P21 and P22 are right too.
9. ESA Jan-24 set 4 2c is the same question as CW P22. The Type 1 and Type 2 terms of the mixed questions are taught in the class 003 and 004 packs; the class 004 pack works only their Type 2 terms and points here for the full solutions.
10. Transcript recognition errors were corrected silently: "extra the parian" → x^n; "5 of d", "5 d", "phi d" → $\varphi(D)$ or $5D$ by context ($D^2 + 5D + 4$); "ascending power set / power sub b" → ascending powers of D; "root 3" → rule 3; "25 D^2 by 60" → $\frac{25D^2}{16}$; "20 by 16" → $\frac{25}{16}$; "particular integraph" → particular integral; "P power n", "b power m", "t square" → D^n, D^m, D^2; "axillary" → auxiliary; "x square plus x square" (T006a 00:03:21) → $x^3 + x^2$.

Coverage

Source	Citations	Slides cited (slide × citations)
S005	77	3×2, 4×27, 5×8, 6×1, 7×6, 8×3, 9×4, 10×12, 11×2, 12×7, 13×3, 14×2
CW	16	
ESA	43	
T005	38	
T006a	26	

Slides 1, 2 and 15 are title and closing slides; slide 3 is the class content. T005 (12.6 min) covers slides 4–9; slides 10–14 (the newer "Type 3: $X = x^m$" format with two problems) are explained in the class 006 problem lecture, whose first 9 min 55 s are Type 3 (tagged T006a here; the class 006 pack cites the same file, for Type 4, as T006b). Bucket numbers are not contiguous (the decks say SESSION 1–11; 007, 008, 010, 012 and 013 do not exist; 009 = session 7, 011 = sessions 8 and 9, 014 = 10, 015 = 11); this deck is SESSION 5.